Candidate's Name: …….....

Signature: ……

Random No.						Personal No.		

(*Do not write your School / Center Name or Number anywhere on this booklet.*)

545/1

CHEMISTRY

Paper 1

April,2026

2 hours

For Scorers' Use Only

Total Weighted Score		
Initials		

CHEMISTRY DEPARTMENT EXAMINATIONS BOARD

Uganda Certificate of Education

INSTRUCTIONS TO CANDIDATES:

This examination paper consists of **two** *sections;* **A** *and* **B**.

It has **six** *examination items*.

Section **A** *has* **two compulsory** *items. Responses to section* **A must** *be written in the spaces provided.*

Section **B** *has* **two** *parts*; **I** *and* **II.** *Respond to only* **one** *item from* **each** *part.*

Responses to section **B must** *be written in the answer booklet*(*s*) *provided.*

Respond to **four** *items in all.*

Responses to each item in section **B** *should start on a fresh page.*

Any additional item(*s*) *responded to will* **not** *be scored.*

For Scorers' Use Only

Item No.	Basic Code	Weighted Score	Scorers' Initials
1	T		
	D		
	E		
2	C		
	X		
	I		
_____(3/4)	P		
	S		
	B		
_____(5/6)	N		
	M		
	Be		
Total Weighted Score			

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SECTION A

Respond to **all** *the items in this section in the spaces provided.*

Item 1

The local council members noted that the residents in the town purchase certain products just after every meal. These products used by the residents are shown in Figure **1.** The local council members have advised the residents to be cautious about the continuous use of these products as they can pose various side effects onto their bodies. Several residents are still confused about the advice from local council members and they have reached out to you for help.



Fig. 1. Products used by the residents.

Task:

(a) State the categories of the products and the function of each product.

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_____

For Scorer's Use Only

Basis Code	T
Score	

(b) Describe the side effect(s) associated with the products and how they are mitigated.

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For Scorer's Use Only

Basis Code	D
Score	

(c) State the;

(i) similarities of the products.

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(ii) Differences between the products.

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For Scorer's Use Only

Basis Code	E
Score	

Item 2

Badru is a famous fish seller in Nakasero market. Recently, Badru's income from selling fish had declined as most of his customers complained that his fish was always rotten and with bad smell.

Badru decided to approach a chemistry teacher in the nearby school for advice. The chemistry teacher told Badru that he should preserve his fish using a compound written as **XY** (*where* **X** *and* **Y** *are not the actual symbols of the elements*), and that the compound can be obtained as shown in the equation below.

$$XOH_{(aq)} + HY_{(aq)} \longrightarrow XY_{(aq)} + H_2O_{(l)}$$

Badru was still confused about the advice from the chemistry teacher and he finally approached you to know more about the compound.

Task:

(a) Explain the nature of the substances **X** and **Y**, and identify them.

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For Scorer's Use Only

Basis Code	C
Score	

(b) Explain the properties of the substances **X** and **Y.**

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(c) Give any other use of the compound **XY**.

_____

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For Scorer's Use Only

Basis Code	X
Score	

(d) Write **one** impact of having a lot of compound **XY** in the environment and how it can be mitigated.

_____

_____

_____

_____

For Scorer's Use Only

Basis Code	I
Score	

SECTION B

Part I

Respond to only **one** *item in this part in the answer booklet*(*s*) *provided.*

Item 3

Nitrogen is an essential nutrient added in fertilizers as it sustains plant growth. The maize farmers' group in a given area experienced poor yields in past seasons due to low levels of nitrogen in the soil.
In response, the government authorized an investor to set up a nitrogen manufacturing plant in the area to be added in required fertilizers.
Aware of the merits and demerits of this manufacturing plant, your school chemistry club wishes to write a sensitization article in daily newspaper. The article is to highlight the production process of nitrogen, the social and economic benefits of the plant to the community and how any side effect(s) is/are to be mitigated. You have been asked to write and present the article.

Task:
Make a write-up you will present in the article.

Item 4

Air is a mixture of different components containing 21% oxygen. Due to wide spread of respiratory illnesses caused by COVID-19, there is an increased demand for oxygen by the patients in hospitals. An investor has set up an oxygen manufacturing plant in an urban center. Figure **2** shows an oxygen manufacturing plant.



Fig. 2. Oxygen manufacturing plant.

However, people in the area have requested to be informed about the process leading to production of oxygen, the social and economic benefits of the manufacturing plant and how any side effect(s) is /are to be mitigated. A radio talk show has been set up to address the request by the people and your school has chosen you as its representative.

Task:
Write details of the key issues you will present during the radio talk show.

Part II

Respond to only **one** *item from this part in the answer booklet*(*s*) *provided.*

Item 5

A youth group in Hoima district has been smuggling crude oil into barrels from the newly constructed East African crude oil pipelines. During the smuggling process, crude oil leaks into the nearby soil.

The crude oil pipeline officials have noted that as the youth continue with this routine, the soil around the pipeline looked abnormal.

The officials expressed worry about the safety of the people around the pipeline and drew attention of the area member of parliament.

In response, the member of parliament has called for a meeting with the youth. You have been chosen to give a sensitization talk. Figure **3** shows crude oil barrels while Figure **4** shows a youth smuggling crude oil.



Fig. 3. Crude oil barrels.



Fig. 4. A youth smuggling crude oil

Task:

Write down the key issues the key issues you will address during the sensitization talk.

Item 6

The environmental authority has noted that the people in the area have been openly dumping waste, plastics, urine and faecal sludge into the only river in the area.

These activities have resulted into release of toxic chemicals into the river hence lowering the water quality.

The environmental authority members have organized a sensitization campaign to find ways of solving the challenges.

The campaign theme is "**Be a water quality champion**." You have been chosen to speak to the rest of the people in the area regarding the theme.

Task:

Make a write-up of the message you will present during the campaign.